Lab 6 PHYS 5394 - Ben Carlson

#1. Show that the covariance of two statistically independent random variables is always zero. First, show that $E[X - E[X]] = 0$ (Hint: $E[X]$ is a constant). Next, use the definition of covariance: $C_{XY} = E[(X - E[X])(Y - E[Y])]$ and the definition of statistical independence (in terms of the joint pdf) along with the above result.

Part 1:

$$\begin{aligned}
E[X - E[X]] &= \int_{-\infty}^{\infty} (X - E[X])\, p_x(x)\, dx \\
&= \int_{-\infty}^{\infty} X p_x(x)\, dx - E[X] \int_{-\infty}^{\infty} p_x(x)\, dx \\
&= E[X] - E[X] \cdot 1 \\
&= 0
\end{aligned}$$

Part 2:

$$\begin{aligned}
C_{XY} &= E[(X - E[X])(Y - E[Y])] \\
&= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (X - E[X])(Y - E[Y])\, p_{xy}(x, y)\, dx dy
\end{aligned}$$

For statistically independent X and Y, $p_{xy}(x, y) = p_x(x)\, p_y(y)$.

$$\begin{aligned}
C_{XY} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (X - E[X])(Y - E[Y])\, p_{xy}(x, y)\, dx dy \\
&= \left(\int_{-\infty}^{\infty} (X - E[X])\, p_x(x)\, dx \right) \left(\int_{-\infty}^{\infty} (Y - E[Y])\, p_y(y)\, dy \right) \\
&= (E[X - E[X]])\ (E[Y - E[Y]]) \\
&= 0 \cdot 0
\end{aligned}$$